DS	**Session**	Arrays	**Question Information**	Level 1 Challenge 27

Problem Description:

Nirobi have given a matrix C of size N x M to Rio.

Also Rio are given position of submatrix as X1, Y1 and X2, Y2 inside the matrix.

Now Rio needs to find the sum of all elements inside that submatrix.

Can you help Rio in completing the task assigned by Nirobi.

Constraints:

$1 \leq T \leq 15$

$1 \leq N, M \leq 103$

$1 \leq C[N][M] \leq 106$

1 <= X1, Y1, X2, Y2 <= M

Input Format:

The first line of input contains an integer T denoting the number of test cases.

The first line of each test case is n and m,n is the number of rows and m is the number of columns.

The second line of each test case contains C[N][M].

The third line contains four value of X1, Y1, X2, Y2. X1, Y1 is the top left cell and X2, Y2 is the bottom right cell.

Output Format:

Print the sum of all elements inside that submatrix.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
1
3 3
91 8 71 14 2 11 6 51 3
1 2 1 3
```

EXPECTED OUTPUT

```
79
```

Test Case 2

INPUT (STDIN)

```
1
3 3
9 18 7 14 2 11 6 51 3
1 2 1 3
```

EXPECTED OUTPUT

```
25
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
int C[m][n];
```

Test Case 2

KEYWORD

```
C[i][j]
```

Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
6	195	33

Code

solution.c

```c
#include <stdio.h>

int main() {
    int t;
    if (scanf("%d", &t) != 1) return 0;
    while (t--) {
        int n, m;
        scanf("%d %d", &n, &m);
        int C[n][m];
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < m; j++) {
                scanf("%d", &C[i][j]);
            }
        }
        int x1, y1, x2, y2;
        scanf("%d %d %d %d", &x1, &y1, &x2, &y2);
        x1--; y1--; x2--; y2--;
        long long sum = 0;
        for (int i = x1; i <= x2; i++) {
            for (int j = y1; j <= y2; j++) {
                sum += C[i][j];
            }
        }
        printf("%lld\n", sum);
    }
    return 0;
}
```

Output

Problems Output Debug Console Terminal Ports powershell

```
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
1
3 3
9 18 7 14 2 11 6 51 3
1 2 1 3
25
PS C:\C Programming>
```